

# Vectors: Components and Operations



## Vector arithmetic

- 1 Let  $\mathbf{u} = \langle 3, -2 \rangle$  and  $\mathbf{v} = \langle -1, 4 \rangle$ . Find:

**a**  $\mathbf{u} + \mathbf{v}$

**b**  $3\mathbf{u} - 2\mathbf{v}$

**c**  $|\mathbf{u}|$

- 2 Find the component form and magnitude of the vector from  $P(1, 3)$  to  $Q(5, -2)$ .
- 3 Find the unit vector in the direction of  $\mathbf{v} = \langle 6, -8 \rangle$ .
- 4 Write the vector of magnitude 12 making an angle of  $150^\circ$  with the positive  $x$ -axis in component form (exact values).

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## Applications

- 5 A plane flies due east at 200 mph while the wind blows due north at 50 mph.
- a** Find the magnitude of the plane's resultant velocity, to one decimal place.
- b** Find the direction of the resultant, as an angle north of east to two decimal places.
- 6 Two forces  $\mathbf{F}_1 = \langle 30, 0 \rangle$  N and  $\mathbf{F}_2 = \langle 0, 40 \rangle$  N act on an object. Find the magnitude of the resultant force.